

Francisco Rau

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PROFILE

Telecommunications engineer, PMP-certified project manager, and Marie Skłodowska-Curie Doctoral Fellow with over 15 years of experience in network engineering, technical leadership, and applied research, with expertise in 5G, MPLS/IP core networks, backhaul, and access networks. Currently completing a Ph.D. in Electronic Engineering at Universitat de València as a Marie Skłodowska-Curie Doctoral Fellow at MaxLinear, focusing on wireless communications, MAC scheduling, and machine learning for telecommunications networks.

EDUCATION

• Universitat de València <i>Ph.D. in Electronic Engineering</i>	Valencia, Spain 2024 – Present
• Universidad de Santiago de Chile <i>M.Sc. in Electrical Engineering</i>	Santiago, Chile 2021 – 2023
• Pontificia Universidad Católica de Chile <i>Diploma in Solar Energy</i>	Santiago, Chile 2017
• Universidad Técnica Federico Santa María <i>B.Sc. in Industrial Engineering</i>	Santiago, Chile 2013 – 2017
• Universidad de Santiago de Chile <i>B.Eng. in Electrical Engineering with a specialization in Electronics and Telecommunications</i>	Santiago, Chile 2005 – 2009

SKILLS

- **Programming Languages:** Python, MATLAB, C++
- **Libraries and Frameworks:** scikit-learn, TensorFlow, Keras, pandas
- **Network Simulation:** ns-3 simulator
- **Platforms:** Linux, Windows, Arduino, Raspberry Pi

PROFESSIONAL EXPERIENCE

- **MaxLinear Hispania S.L.** Valencia, Spain
Researcher in Communication Systems — Full-time March 2024 – Present
 - **Role:** Ph.D. researcher investigating efficient MAC scheduling for ultra-dense wireless networks based on IEEE 802.11 standards.
- **Entel S.A.** Santiago, Chile
Senior Project Manager for Services and Networks — Full-time July 2015 – March 2024
 - **Role:** Project management across Entel Chile’s data, voice, and packet core networks, coordinating with international vendors such as Huawei, Cisco, and Ericsson.
 - **Achievements:** Successfully delivered highly complex telecommunications projects, including 5G NSA, SDN networks, MPLS/IP core expansion, and CDN infrastructure supporting Google, Facebook, Netflix, Microsoft, and Amazon.
- **Entel S.A.** Santiago, Chile
Project Manager for Backhaul and Access Networks — Full-time April 2013 – June 2015
 - **Role:** Project management of access and backhaul networks using fiber-optic (FO), microwave (MW), and satellite technologies for 4G/LTE mobile sites and corporate customers.
 - **Achievements:** Successfully delivered the transport network and customer access nodes for the Chilean Ministry of Health in under one year.
- **ASETA, Aso. de Telecom. de la C. Andina** Quito, Ecuador
FTTH Consultant — Full-time November 2011 – October 2012
 - **Role:** Consulting and advisory services on FTTH networks for CNT EP (National Telecommunications Corporation).
 - **Achievements:** Redesigned the project implementation approach, reducing deployment timelines to under six months. Developed procedures and delivered training related to the transition from copper to fiber-optic technology.

- **Entel S.A.** Santiago, Chile
Project Engineer — Full-time *September 2009 – November 2011*
 - **Role:** Management and coordination of GPON, microwave (MW), and fiber-optic (FO) projects supporting the expansion of 3G/4G mobile and fixed networks.
 - **Achievements:** Technical lead for a large-scale commercial GPON/FTTB deployment, the only project in LATAM to connect approximately 1,000 buildings in less than one year.
- **Tu Ves S.A.** Curacaví, Chile
Satellite Station Engineer — Full-time *March 2009 – September 2009*
 - **Role:** Responsible for operating the satellite digital TV platform (headend and uplink), including maintenance, fault diagnosis, and supervision of reception, encryption, encoding, and satellite transmission systems.
- **Telmex S.A.** Santiago, Chile
Operations and Maintenance Engineer — Full-time *January 2008 – April 2008*
 - **Role:** Supported operations and maintenance activities at the satellite ground station, including C- and Ku-band uplink and downlink systems.

ACADEMIC EXPERIENCE

- **Optical and Wireless Sensors Networks for 6G scenarios (OWIN6G)** Valencia, Spain
Doctoral Researcher — Full-time *March 2024 – Present*
 - **Role:** Marie Skłodowska-Curie Doctoral Fellow within the European HORIZON-MSCA-2022-DN-01 project Optical and Wireless Sensors Networks for 6G scenarios (OWIN6G) – <https://owin6g.eu/>.
- **Czech Technical University in Prague (CTU)** Prague, Czech Republic
Visiting Researcher — Full-time *September 2026 – Present*
 - **Role:** Research on digital twin modeling for ultra-dense Wi-Fi networks.
- **Instituto de Telecomunicações** Aveiro, Portugal
Visiting Researcher — Full-time *February 2025 – March 2025*
 - **Role:** Research on incremental learning techniques for network traffic classification.
- **DSDC Group** Valencia, Spain
Visiting Researcher — Full-time *October 2024 – December 2024*
 - **Role:** Research on AI-enabled embedded systems for real-time data processing, machine learning, and hardware acceleration.
- **Center for Multidisciplinary Research in Telecommunications Technologies (CIMTT)** Santiago, Chile
Researcher — Part-time *March 2021 – July 2023*
 - **Role:** Research on network traffic prediction using neural networks and machine learning, with applications to telecommunications operators.
- **Duoc UC** Santiago, Chile
Engineering Lecturer — Part-time *March 2018 – July 2018*
 - **Role:** Lecturer at the School of Engineering in the Electrical and Industrial Automation programs, teaching Applied Physics, Industrial Mechanisms, and Human Resources Management.
- **ASETA, Aso. de Telecom. de la C. Andina** Quito, Ecuador
Instructor — Part-time *March 2012 – August 2012*
 - **Role:** Training CNT EP technical personnel in the design and deployment of optical distribution networks (ODNs) and the installation of optical line terminals (OLTs) using GPON technology for Ecuador's pilot and national deployment plans.

ACADEMIC PUBLICATIONS

- **Francisco Rau**, Ismael Soto, Pablo Adasme, David Zabala-Blanco, and Cesar A. Azurdia-Meza, “Network Traffic Prediction Using Online-Sequential Extreme Learning Machine”, *3rd South American Colloquium on Visible Light Communications (SACVLC 2021)*, Toledo, Brazil, 11–12 November 2021. DOI: 10.1109/SACVLC53127.2021.9652247.
- **Francisco Rau**, Ismael Soto, and David Zabala-Blanco, “Forecasting Mobile Network Traffic based on Deep Learning Networks”, *IEEE Latin-American Conference on Communications (LATINCOM 2021)*, Santo Domingo, Dominican Republic, 17–19 November 2021. DOI: 10.1109/LATINCOM53176.2021.9647788.
- Raul Zamorano-Illanes, María Constanza Estela, Ismael Soto, Muhammad Ijaz, and **Francisco Rau**, “MIMO QAM indoor VLC using polar codes for low-cost emitters and FPGA receiver”, *4th West Asian Symposium on Optical and Millimeter-Wave Wireless Communications (WASOWC 2022)*, Tabriz, Iran, 12–13 May 2022. DOI: 10.1109/WASOWC54657.2022.9798443.
- Raul Zamorano-Illanes, Ismael Soto, Wilson Alavia, Verónica García, Pablo Adasme, and **Francisco Rau**, “Remote detection of COVID-19 using 5G and AI”, *13th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP 2022)*, pp. 395–400, Porto, Portugal, 20–22 July 2022. DOI: 10.1109/CSNDSP54353.2022.9907995.

- **Francisco Rau**, Ismael Soto, David Zabala-Blanco, Cesar Azurdia-Meza, Muhammad Ijaz, Sunday Ekpo, and Sebastian Gutierrez, “A Novel Traffic Prediction Method Using Machine Learning for Energy Efficiency in Service Provider Networks”, *Sensors*, 2023, 23(11), 4997. DOI: 10.3390/s23114997.
- **Francisco Rau**, Carlos Herranz, Iñaki Val, Petia Georgieva, and Joaquin Perez, “A Novel Flow-Based Online Network Traffic Classification Using Machine Learning”, *TechRxiv*, 2025. DOI: 10.36227/techrxiv.175459507.75164419/v1.
- **Francisco Rau**, Carlos Herranz, Iñaki Val, and Joaquin Perez, “LSTM-Based Traffic Prediction for 5G Mobile Network Operator”, *2025 International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, Split, Croatia, 18–20 September 2025. DOI: 10.23919/SoftCOM66362.2025.11197406.
- **Francisco Rau**, Luis Miguel Giraldo, Carlos Herranz, Joaquin Perez, Iñaki Val, and Raimundo Garcia, “Implementing a Traffic Classifier on Embedded SoC Systems with Deep Learning Processor Unit”, *2025 International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, Split, Croatia, 18–20 September 2025. DOI: 10.23919/SoftCOM66362.2025.11197409.
- **Francisco Rau**, Petia Georgieva, Carlos Herranz, Iñaki Val, and Joaquin Perez, “Incremental Learning in Network Traffic Management: A Fixed-Representation Rehearsal Approach”, *15th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP 2026)*, Edinburgh, United Kingdom, 15–17 July 2026. DOI: 10.1109/CSNDSP68462.2026.11654486.
- Luis Miguel Giraldo, Joaquin Perez, Raimundo Garcia, Atiyeh Pouralizadeh, Volker Jungnickel, **Francisco Rau**, and Carlos Herranz, “A SDN-Based Access Control Framework for Powerline Communications Networks”, *15th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP 2026)*, Edinburgh, United Kingdom, 15–17 July 2026. DOI: 10.1109/CSNDSP68462.2026.11654376.

RESEARCH DATASET

- **Francisco Rau**, Carlos Herranz, Iñaki Val, and Joaquin Perez, “VLC Data: A Multi-Class Network Traffic Dataset Covering Diverse Applications and Platforms”, *EU Open Research Repository, Zenodo*, April 2025. DOI: 10.5281/zenodo.15121418.

CERTIFICATIONS AND TRAINING

- Project Management Professional (PMP)[®] - Credential #2746245, renewal due 30 January 2029
- Research Colloquium, NextGCom2025 – Greece, October 2025
- Training – The Double Nature of ML/AI in Cybersecurity, Albania, June 2025
- Python Workshop for Data Science, Huawei Enterprise Service – September 2022
- Innovation Management Certified Professional (IMCP) – March 2021
- Advanced Programming Tools in Python for Data Processing, PUC – November 2020
- Scrum Foundation Professional Certified (SFPC) #42152609 – July 2020
- Proof of Concept Testing for Huawei NE-9000 Router Series, Huawei, Beijing, China – July 2018
- iManager U2000 DSLAM MA5616 Series O&M Training, Huawei, Brazil – March 2014
- DSLAM SmartAX MA5616 Series 2nd Line Maintenance Training, Huawei, Brazil – March 2014
- LTE for Entel Technical Area, University of Chile – August 2013
- Fault Tolerance in Industrial Wireless Sensor Networks, UACH, Valdivia – November 2008

AWARDS AND RECOGNITION

- Marie Skłodowska-Curie Doctoral Fellow, 2024
- Ranked 1st in the M.Sc. in Electrical Engineering program, USACH – November 2023
- Top 10 graduate in Industrial Engineering for academic performance, UTFSM – November 2017
- Ranked 1st in the B.Eng. in Electrical Engineering program, USACH – September 2009

LANGUAGES

- Spanish — Native
- English — Upper-intermediate proficiency